

Technical Assignment: Event Registration Mini App

Overview

Build a small full-stack Event Registration application.

This task is designed to assess your ability to build a working frontend, backend API, MongoDB integration, basic form flow, filtering logic, and simple analytics-style event tracking.

The assignment should be completed within **72 hours**.

Objective

Create a mini application where users can:

1. View a list of upcoming events.
 2. Search and filter events.
 3. Open an event details view.
 4. Register for an event using a form.
 5. Store registrations in MongoDB.
 6. Track basic user actions in the app.
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Tech Stack

Please use:

Frontend

- React.js or Next.js

Backend

- Node.js
- Express.js

Database

- MongoDB

You may use JavaScript or TypeScript.

Functional Requirements

1. Event Listing Page

Create a page that displays a list of events.

Each event should have:

- Event name
- Date
- Category
Example: Workshop, Seminar, Hackathon, Webinar
- Location or mode
Example: Noida, Online, Hybrid
- Short description
- Available seats

The page should support:

- Search by event name
- Filter by category
- Filter by location/mode

2. Client-Side and Server-Side Filtering

Implement both approaches:

A. Client-Side Filtering

Fetch all events once and filter them on the frontend using React state.

B. Server-Side Filtering

Create an API that supports filters through query parameters.

Example:

```
GET /api/events?category=Workshop&search=react&mode=Online
```

In the UI, provide a simple toggle or two buttons:

- Client-side filter
- API-side filter

This is important because it tests both frontend state handling and backend query handling.

3. Event Details Page or Modal

When a user clicks an event, show more details.

This can be either:

- A separate page, or
- A modal

Show:

- Event name
 - Full description
 - Date
 - Category
 - Location/mode
 - Available seats
 - Register button
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4. Registration Form

Create a registration form for an event.

The form should collect:

- Name
 - Email
 - Phone number
 - College/company
 - Source
- Example: LinkedIn, WhatsApp, Instagram, Email, Direct

Validation rules:

- Name is required.
- Email should be valid.
- Phone number should be valid.
- The same email should not be able to register twice for the same event.
- Registration should not be allowed if seats are full.

After successful registration:

- Save the registration in MongoDB.
 - Reduce available seats or update registration count.
 - Show a success message.
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5. Backend APIs

Create the following APIs:

```
GET /api/events
GET /api/events/:id
POST /api/events
POST /api/events/:id/register
GET /api/events/:id/registrations
```

The backend should include:

- MongoDB models
 - Proper API response structure
 - Basic validation
 - Error handling
 - CORS setup
 - Environment variable usage
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6. Basic Analytics Tracking

Implement a simple analytics tracking utility.

You can either use GA4/GTM or create a mock function like:

```
trackEvent("event_card_clicked", {
  eventId,
  eventName,
  category
});
```

Track at least these actions:

```
event_list_viewed
event_search_performed
event_filter_applied
event_card_clicked
registration_submitted
registration_success
registration_failed
```

You may log these events in the browser console. Bonus points if you store them in MongoDB.

In the README, briefly explain:

- Which events you tracked
 - Where they are triggered
 - What payload you send
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7. Simple Dashboard

Create a basic dashboard section or page showing:

- Total events
- Total registrations
- Registrations per event
- Recent registrations

This can be a simple table/card layout. Charts are optional.

8. Seed Data

Add sample data for at least:

- 8 events
- 3 categories
- 10 sample registrations

You may provide either:

- A seed script, or
 - Clear instructions in the README to insert sample data
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9. Deployment

Deployment is preferred but not mandatory.

If deployed, share:

- Frontend URL
- Backend URL

If not deployed, the app should run locally with clear instructions.

Submission Deliverables

Please submit:

1. GitHub repository link
 2. README.md with setup instructions
 3. Screenshots or short demo video
 4. Any deployed links, if available
 5. Notes on any errors/issues faced and how you debugged them
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Evaluation Criteria

1. Full-Stack Integration — 30%

- Frontend communicates correctly with backend
- Data is fetched from MongoDB
- Registration flow works end-to-end
- API errors are handled properly

2. Frontend Quality — 20%

- React components are structured cleanly
- State is handled correctly
- Search/filter UI works properly
- Form validation is clear
- UI is responsive and understandable

3. Backend Quality — 20%

- APIs are well structured
- MongoDB schemas are sensible
- Server-side filtering works
- Duplicate registration and full-seat cases are handled
- Code is readable and modular

4. Analytics and Product Thinking — 15%

- Important user actions are tracked
- Event payloads are meaningful
- Candidate explains how tracking can help improve the product

5. Documentation and Debugging — 15%

- README is clear
 - Setup steps work
 - Known issues are documented
 - Candidate explains debugging steps if something fails
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Bonus Points

Bonus points for:

- TypeScript
 - Next.js
 - Deployment
 - Debounced search
 - Basic authentication for dashboard
 - CSV export of registrations
 - GA4 or GTM integration
 - Clean Git commits
 - Good mobile responsiveness
 - Use of AI coding assistants with a short note explaining what was AI-assisted
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Timeline

You have **72 hours** from the time of receiving this assignment.

Please focus on a clean, working implementation rather than adding too many unfinished features.